

Maliha Tariq

Software Engineer

malihatariqq@gmail.com | +971 58 123 3911 | United Arab Emirates | maliha-folio.vercel.app | linkedin.com/in/malihatariqq

SUMMARY

Software Engineering graduate specialising in applied machine learning, from deep learning model design (PyTorch, wav2vec2) to shipping AI features inside production-style web and mobile apps, with internship experience in frontend development, UAT, backend automation and agile delivery.

EDUCATION

Bachelor of Software Engineering (Honours) 06/2026

Monash University Malaysia | CGPA: 3.7 | Best Final Year Project Award (2026)

EXPERIENCE

IT Project Management Intern, Fireworks Solutions International 11/2024 – 02/2025

- Developed backend automation scripts that reduced manual workload, supporting API workflows and structured data pipelines.
- Translated high-fidelity Figma prototypes into responsive, pixel-accurate web interfaces, collaborating directly with the design lead.
- Authored and executed UAT scripts to validate functionality and confirm acceptance criteria.
- Assisted project planning, milestone tracking, and cross-team coordination in a fast-paced, client-facing agile delivery environment, proactively raising blockers and adapting to shifting requirements.

PROJECTS

SEMS2: Student Experience Management System 10/2025

SvelteKit · TypeScript · Tailwind · Supabase (PostgreSQL, Auth, Realtime)

- Built a role-based (Student / Organiser / Admin) full-stack web app used in production by Monash clubs and societies for the full event lifecycle: discover → register → feedback → certificates.
- Implemented Google OAuth restricted to institutional accounts, real-time data sync, and CSV/XLS export pipelines for admin reporting.
- Set up automated testing (Vitest, Playwright) and a defined release cadence with UAT.

NutriTrack: Clinical Nutrition Insights App 06/2025

Kotlin · Jetpack Compose · Room DB · Retrofit · Google Gemini (GenAI)

- Built an Android app applying MVVM architecture with Room DB for persistent, normalised data, integrating the FruityVice API via Retrofit for live nutrition data.
- Integrated Google Gemini (GenAI) to generate contextual, personalised nutrition guidance from raw clinical scores to shipped LLM output directly into a user-facing feature.
- Implemented multi-user authentication and a secure admin analytics dashboard (HEIFA scoring).

Astone: E-Commerce Platform 07/2024

React.js · Django · PostgreSQL · Figma · Vercel

- Built the front-end in React.js and backend services in Django, connected to PostgreSQL for user and inventory data management.
- Deployed and hosted the application on Vercel.

Phoneme-Informed Speech Impairment Classifier: Final Year Project 06/2026

Python · wav2vec2 · PyTorch · librosa · TORGO Dataset

- Engineered a feature-injection pipeline adding pitch and MFCC acoustic features into a pretrained wav2vec2 model, cutting Phoneme Error Rate by 72% (0.527 → 0.147) across 15 dysarthric speakers.

- Validated results statistically (Wilcoxon signed-rank test, $p = 3.05 \times 10^{-5}$) and ran a 5-configuration ablation study to isolate each feature's impact — rigorous, reproducible model evaluation.

TECHNICAL SKILLS

ML / AI: PyTorch, wav2vec2, librosa, Google Gemini (GenAI) API integration & model evaluation

Languages: Python, Java, JavaScript, TypeScript, Kotlin, PHP

Frontend: React.js, SvelteKit, Tailwind CSS, Jetpack Compose, HTML & CSS, Figma

Backend / Data: Django, Retrofit, PostgreSQL (Supabase), MySQL, Oracle

Tools & Testing: GitHub, Vitest, Playwright, Agile/Scrum, Trello, Jira